

Example 1 — Integers and identifiers

Create a file called scanner.l and paste the following.

```
%option noyywrap

%{
#include <stdio.h>
%}

%%

[0-9]+          { printf("INTEGER: %s\n", yytext); }

[a-zA-Z][a-zA-Z0-9]*  { printf("IDENTIFIER: %s\n", yytext); }

[ \t\n]+        { /* ignore whitespace */ }

.                { printf("UNKNOWN: %s\n", yytext); }

\n              {return 0;}

%%

int main()
{
    yylex();
    return 0;
}
```

Test input

```
age 25
student1 2026
total 500
```

Example 2 — Keywords and operators

```
%option noyywrap
```

```
%{  
#include <stdio.h>  
%}
```

```
%%
```

```
"int"          { printf("KEYWORD: INT\n"); }  
"if"           { printf("KEYWORD: IF\n"); }  
"else"         { printf("KEYWORD: ELSE\n"); }  
"return"       { printf("KEYWORD: RETURN\n"); }
```

```
[0-9]+         { printf("INTEGER: %s\n", yytext); }
```

```
[a-zA-Z][a-zA-Z0-9]* { printf("IDENTIFIER: %s\n", yytext); }
```

```
"="           { printf("ASSIGN: %s\n", yytext); }  
"+"          { printf("PLUS: %s\n", yytext); }  
"-"          { printf("MINUS: %s\n", yytext); }  
"*"          { printf("MULTIPLY: %s\n", yytext); }  
"/"          { printf("DIVIDE: %s\n", yytext); }
```

```
[\t\n]+       { }
```

```
.             { printf("UNKNOWN: %s\n", yytext); }
```

```
%%
```

```
int main()  
{  
    yylex();  
    return 0;  
}
```

Example 3 — Scan a small program

```
">="    { printf("GREATER_EQUAL: %s\n", yytext); }
```

```
"("     { printf("LEFT_PAREN: %s\n", yytext); }
```

```
")"     { printf("RIGHT_PAREN: %s\n", yytext); }
```

```
"{"     { printf("LEFT_BRACE: %s\n", yytext); }
```

```
"}"     { printf("RIGHT_BRACE: %s\n", yytext); }
```

```
","     { printf("SEMICOLON: %s\n", yytext); }
```

Example 4: Create a scanner.l file and add the following code

```
/* Definitions */
%{
#include <stdio.h>

int keywords = 0;
int identifiers = 0;
int numbers = 0;
int operators = 0;
int special_characters = 0;
}%

/* Rules and Actions */
%%
"int"|"float"|"char"|"if"|"else"|"while"|"return" { keywords++; }

[a-zA-Z_][a-zA-Z0-9_]*      { identifiers++; }

[0-9]+      { numbers++; }

"+"|"-"|"*"|"/"|"="|"=="|"!="|"<"|">"      { operators++; }

[{}();,\[\]]      { special_characters++; }

[\t\n]+      { /* ignore whitespace */ }

.      { printf("Unknown character: %s\n", yytext); }

%%

int yywrap()
{
    return 1;
}

int main()
{
    yyin = fopen("test.txt", "r");

    if (yyin == NULL)
    {
        printf("Could not open test.txt\n");
        return 1;
    }
}
```

```

}

yylex();

printf("\nLexical Analysis Results:\n");
printf("-----\n");
printf("Keywords      : %d\n", keywords);
printf("Identifiers    : %d\n", identifiers);
printf("Numbers        : %d\n", numbers);
printf("Operators      : %d\n", operators);
printf("Special characters : %d\n", special_characters);

fclose(yyin);

return 0;
}

```

Create a test.txt file and add the following code

```

int age = 20;
float salary = 50000;

if (age > 18) {
    salary = salary + 1000;
}

```